



Proposed CEA Link — December 2026

The existing 20-line link, widened to 24 people — the same turns, the same shape of week, one more cover cycle

Prepared 8 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules · figures on this page are computed, not typed

PROPOSAL

Same Turns

Today's link, widened to 24

CODE **ST-24-B7**

FINGERPRINT **d15e1b74**

BUILT FROM duty table B · seed 7 · 24 lines · 4 cover weeks

LINEAGE Family ST — the existing 20-line link kept in its own shift times and week shapes. Compare with "Five Shift Times" (23 Aug 2026, family FST) and the workspace's own December default.

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **1** fatigue factor present (today: 5)

15 of 15

shift times are today's

every time on the sheet is one people already work — nothing new to learn

6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

18 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20)

4 cover weeks

lines 1, 7, 13, 19

evenly spaced six lines apart — today's sit at 1, 7, 12, 17

2h 33m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by hand and it was not produced by the workspace's generator, which slides a person's start time through the week. Instead the December duty table was written in today's shift times, and a search then arranged those duties across 24 lines — over a million candidate arrangements — with the workspace's own hard-limit, rest, contract, fatigue and line-order modules scoring every one, plus a measure of how closely each resembled today's roster. Page 7 says how, and how to reproduce it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 5 and 6 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
2	RD	RD	RD	08:00 16:30	08:00 16:30	08:00 16:30	06:20 14:50	34h 00m	34h 00m	4
3	07:15 15:45	08:00 16:30	RD	RD	08:00 16:30	08:00 16:30	06:20 14:00	33h 10m	41h 40m	4
4	07:15 15:45	RD	RD	06:20 13:45	06:20 13:45	06:20 13:45	RD	22h 15m	30h 45m	3
5	RD	13:30 22:00	13:30 22:00	13:30 22:00	13:30 22:00	RD	14:30 22:00	41h 30m	41h 30m	5
6	14:30 23:25	15:15 23:55	15:15 23:55	RD	RD	15:15 23:55	14:45 23:55	35h 10m	44h 05m	4
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
8	RD	RD	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	RD	32h 00m	32h 00m	4
9	RD	06:20 13:45	06:20 13:45	06:20 13:45	06:20 13:45	RD	RD	29h 40m	29h 40m	4
10	RD	06:20 14:20	08:00 16:30	08:00 16:30	06:20 14:20	06:20 14:20	RD	41h 00m	41h 00m	5
11	RD	15:15 23:55	15:15 23:55	15:15 23:55	15:15 23:55	15:15 23:55	RD	43h 20m	43h 20m	5
12	RD	15:15 23:55	15:15 23:55	15:15 23:55	15:15 23:55	RD	14:45 23:55	43h 50m	43h 50m	5
13	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
14	11:00 19:30	RD	RD	14:00 22:30	14:00 22:30	14:00 22:30	12:00 20:00	33h 30m	42h 00m	4
15	11:00 19:30	RD	RD	15:15 23:55	15:15 23:55	15:15 23:55	14:30 22:00	33h 30m	42h 00m	4
16	14:30 23:25	14:00 22:30	RD	RD	14:00 22:30	14:00 22:30	14:45 23:55	34h 40m	43h 35m	4
17	RD	08:00 16:30	08:00 16:30	RD	RD	08:00 16:30	06:20 14:50	34h 00m	34h 00m	4
18	07:15 15:45	06:20 13:45	06:20 13:45	RD	RD	06:20 13:45	08:00 16:30	30h 45m	39h 15m	4
19	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
20	RD	RD	13:30 22:00	13:30 22:00	13:30 22:00	13:30 22:00	RD	34h 00m	34h 00m	4
21	RD	08:00 16:30	08:00 16:30	08:00 16:30	08:00 16:30	RD	06:20 14:50	42h 30m	42h 30m	5
22	07:15 15:45	06:20 14:20	06:20 14:20	06:20 14:20	RD	RD	08:00 16:30	32h 30m	41h 00m	4
23	14:30 23:25	13:30 22:00	14:00 22:30	RD	RD	13:30 22:00	14:00 22:30	34h 00m	42h 55m	4
24	13:00 21:00	14:00 22:30	14:00 22:30	14:00 22:30	RD	RD	14:45 23:55	34h 40m	42h 40m	4
Cover	10 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	35h 00m	38h 34m	4.20

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines

Same feel, more people

What the search was told to keep, and what the December headcounts made it change

Kept — the shape of a week

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	18 of 20
Weeks mixing early and late turns	7	0
Rest-day breaks that are two days or more	13 of 17	20 of 24
Single rest days between duties (not a 48h break)	4	4
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	6×4, 14×5
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 13, 19
Distinct shift times (all already on today's roster)	18	15
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	10 of 20 lines

Changed — the December headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	14	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	14	owner's figure for December, leaning late for events
Sunday	8	10	owner's figure for December
Opening at 06:20 (07:15 Sunday)	4	4	unchanged
Through to the 23:55 close, weekday	2	3	three to close
Through to the close, Saturday	2	4	four on a Saturday
Still on duty at 22:00	4	5	five are enough from 22:00

The duty table, in today's times

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:35 7h 15m · Early	4			4		
06:20–13:45 7h 25m · Early	1			2 +1		
06:20–14:00 7h 40m · Early		1			1	
06:20–14:20 8h 00m · Early	2			2		
06:20–14:50 8h 30m · Early		3			3	
07:15–15:45 8h 30m · Early			3			4 +1
08:00–16:30 8h 30m · Early	2			3 +1	2 +2	
08:30–16:30 8h 00m · Early			4			4
11:00–19:30 8h 30m · Late	1			-1		2 +2
12:00–20:00 8h 00m · Late		1			1	
13:00–21:00 8h 00m · Late			1			1
13:30–21:00 7h 30m · Late		4			4	
13:30–22:00 8h 30m · Late	1			2 +1		
14:00–22:30 8h 30m · Late	2			2	1 +1	
14:30–22:00 7h 30m · Late		2			2	
14:30–23:25 8h 55m · Late			3			3
14:45–23:55 9h 10m · Late		2			4 +2	
15:15–23:55 8h 40m · Late	2			3 +1		

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the extra duties land where they do: a weekday needs 6,980 minutes and a Saturday 7,100 for 5 weekdays + 1 Saturday to total exactly $20 \times 35h = 42,000$ — an equality, so the table is one of the few combinations of today's turn lengths that reaches it to the minute. The 06:20–13:35 turn is folded into 06:20–13:45 (ten minutes, one fewer time to remember); Sunday gains an 11:00–19:30 middle turn from the weekday vocabulary.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

Monday to Friday window 06:20–23:55 · one row per distinct weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2
PROPOSED		4	4	7	7	7	7	7	9	9	10	10	7	7	7	7	7	5	3
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2
PROPOSED		4	4	6	6	6	6	7	7	13	10	10	8	8	8	7	7	5	4
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3
PROPOSED			4	4	4	4	6	6	7	10	10	6	6	6	6	4	3	3	3
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. Sunday: 5 December movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

Reading it. The proposal raises the whole day rather than reshaping it, which is what the December brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (85.75h) sit on top as RDW, as they do today. Individual weeks range 22h 15m–43h 50m; only the average is the contract.

December design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
Four on at the open, every day	✓ 4 weekday · 4 Saturday · 4 Sunday	
Three through to the close; four on a Saturday	✓ 3 weekday · 4 Saturday · 3 Sunday	
Five still on duty at 22:00	✓ 5 weekday · 5 Saturday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	✓ 14 · 10	
Four cover weeks, evenly spread	✓ lines 1, 7, 13, 19 — gaps 6, 6, 6, 6	whole weeks, never scattered days
About 4.2 days a week worked, Mon-Sat	✓ 4.20 over the 20 working lines	
No :05 or :10 times except the open and close	✓ none	today's times already obey this
Late turns slightly shorter than most earlies	× weekday not met (lates 8h30–8h40, earlies 7h25–8h30) · Saturday not met (lates 7h30–9h10, earlies 7h40–8h30) · Sunday not met (lates 8h00–8h55, earlies 8h30–8h30)	incompatible with keeping today's times: meeting it needs 9h+ earlies or 15 duties a day — flagged for decision, below

Two things to settle before it is frozen

Late-turn length. The one December preference this proposal does not meet is the owner's own — lates slightly shorter than most earlies. Today's link has it the other way round (a 15:15–23:55 late is 8h40, a 06:20–14:20 early 8h00) and the brief was to keep today's times; meeting both is arithmetically impossible at 14 duties a day, because the day's minutes are fixed by the contract. The choice is between today's clock times and shorter lates paid for by longer earlies (the workspace default does this, with 9h25 earlies). **Sunday's finish** — five December movements fall after 23:25; the proposal inherits today's window deliberately rather than deciding it.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). Δ present · ✓ clear · ● standing, a property of the operation · – not applicable. Nathan assesses against this list.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	– –	– –
FF2	Early shift starting 05:00–07:00 Time of day	● 24	● 24
FF3	Very early shift starting before 05:00 Time of day	✓ 0	✓ 0
FF4	Very early shift (before 05:00) over 8h Duty length	– –	– –
FF5	Day shift over 12h Duty length	✓ 0	✓ 0
FF7	Early shift over 10h Duty length	✓ 0	✓ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	✓ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	✓ 12 ✓ 9 as rostered	✓ 10 ✓ 9 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	✓ 0	✓ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	✓ 4
MRSF	More than 12 consecutive day shifts Cumulative	✓ 9	✓ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	✓ 51
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	Δ 9	✓ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	– –	– –
FF17	Backward rotating pattern (definition to confirm) Circadian	✓ 10 backward / 12 forward	✓ 13 backward / 13 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	● 20-line rotation · typically 4h 0m a week	● 24-line rotation · typically 2h 33m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6	Δ 1
FF6	Night shift over 10h Night shifts	– –	– –
FF8	Less than 2 days rest after consecutive nights Night shifts	– –	– –
FF9	Less than 14h rest in any 24h for night shifts Night shifts	– –	– –
FF12	Only one day rest after night shifts Night shifts	– –	– –
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	– –	– –
FF16	More than 3 consecutive night shifts over 8h Night shifts	– –	– –
FF20	First night shift Night shifts	– –	– –
FF13	Less than 12h rest in any 24h Intervals between duties	✓ 0	✓ 0

Present: 1 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

How it was chosen, and how to load it

Reproducible: the same table, seed and objective give the same grid on any machine

How it was chosen

1 • The table. Every possible December duty table was enumerated from today's shift times — every count of each turn, on each day class — and kept only if it opened with four, closed with three (four on Saturday), kept five on at 22:00, and totalled exactly 42,000 Mon–Sat minutes. 81 tables survive. Two weekday shapes were carried forward — table A with three 11:00 middles, table B with a third 08:00 and today's 13:30 turn — and every finished design was scored against the December demand curve; B, used here, follows a weekday's curve about as evenly as today's roster does (57.7 today, 58.5 here, lower is more even), where A follows it less well (69.5) — the three 11:00 middles put nine people on the quietest part of the afternoon.

2 • The grid. Simulated annealing over the 24-line grid: 60,000 candidate arrangements per restart, four restarts from different random starting points per run, three seeded runs per table — over a million candidates in all — then a steepest-descent polish of every same-day swap and every pair of lines. Every move keeps each day's duties exactly as the table says — a swap between two lines on one day, or two whole lines changing places — so the coverage, the contract and the headcounts are true by construction and only the SHAPE is searched. Cover weeks stay at lines 1, 7, 13 and 19 throughout.

3 • The judge. Every candidate was scored by the workspace's own modules — `runDesignChecks`, `assessFatigue`, `assessHardLimits`, `scoreOrder` — with a rest under 12h or a run past six days costing more than everything else combined, fatigue factors next, and then the feel terms: a week on more than one turn, a single rest day between duties, fewer than four or more than five days, and the week-to-week step. Weekends off and long weekends were rewarded. Last, the app's line-order optimiser was offered the result and kept only where it improved that score.

Paste it into the workspace

Links → Import, then paste the block on page 8. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied as `proposal-import.txt` (tab-separated, pastes directly) and `proposal.json` (the app's own format).

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE-TURN	STEP	WK FIT	SCORE
ST-24-B7 · d15e1b74 — Same Turns (this proposal)	6	1	6	18/20	2h 33m	58.5	8268
ST-24-A21 · 235ecdd7 — runner-up	6	1	6	18/20	1h 47m	69.5	6742
ST-24-A13 · 22966914	6	1	5	17/20	2h 31m	69.5	10806
ST-24-B21 · 8a3198a8	6	2	6	18/20	3h 8m	58.5	11528
ST-24-A7 · 320073c5	6	2	6	19/20	3h 31m	69.5	11566
ST-24-B13 · adaa379c	6	2	5	18/20	2h 26m	58.5	11826
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	41.1	2192436
BB-24-D7 · 0f14abce — By the Book (a sibling proposal)	6	0	6	12/20	2h 52m	41.1	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	56	—

How the winner was picked: rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December traffic curve (lower is better; today's link scores what it scores). Two candidates tied on every rule; the fit decided it, and the runner-up is named so it can be asked for. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

1	SP	SP	SP	SP	SP	SP	SP
2	RD	RD	RD	08:00-16:30	08:00-16:30	08:00-16:30	06:20-14:50
3	07:15-15:45	08:00-16:30	RD	RD	08:00-16:30	08:00-16:30	06:20-14:00
4	07:15-15:45	RD	RD	06:20-13:45	06:20-13:45	06:20-13:45	RD
5	RD	13:30-22:00	13:30-22:00	13:30-22:00	13:30-22:00	RD	14:30-22:00
6	14:30-23:25	15:15-23:55	15:15-23:55	RD	RD	15:15-23:55	14:45-23:55
7	SP	SP	SP	SP	SP	SP	SP
8	RD	RD	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20	RD
9	RD	06:20-13:45	06:20-13:45	06:20-13:45	06:20-13:45	RD	RD
10	RD	06:20-14:20	08:00-16:30	08:00-16:30	06:20-14:20	06:20-14:20	RD
11	RD	15:15-23:55	15:15-23:55	15:15-23:55	15:15-23:55	15:15-23:55	RD
12	RD	15:15-23:55	15:15-23:55	15:15-23:55	15:15-23:55	RD	14:45-23:55
13	SP	SP	SP	SP	SP	SP	SP
14	11:00-19:30	RD	RD	14:00-22:30	14:00-22:30	14:00-22:30	12:00-20:00
15	11:00-19:30	RD	RD	15:15-23:55	15:15-23:55	15:15-23:55	14:30-22:00
16	14:30-23:25	14:00-22:30	RD	RD	14:00-22:30	14:00-22:30	14:45-23:55
17	RD	08:00-16:30	08:00-16:30	RD	RD	08:00-16:30	06:20-14:50
18	07:15-15:45	06:20-13:45	06:20-13:45	RD	RD	06:20-13:45	08:00-16:30
19	SP	SP	SP	SP	SP	SP	SP
20	RD	RD	13:30-22:00	13:30-22:00	13:30-22:00	13:30-22:00	RD
21	RD	08:00-16:30	08:00-16:30	08:00-16:30	08:00-16:30	RD	06:20-14:50
22	07:15-15:45	06:20-14:20	06:20-14:20	06:20-14:20	RD	RD	08:00-16:30
23	14:30-23:25	13:30-22:00	14:00-22:30	RD	RD	13:30-22:00	14:00-22:30
24	13:00-21:00	14:00-22:30	14:00-22:30	14:00-22:30	RD	RD	14:45-23:55

Prepared 8 September 2026. Every figure in this document was computed by the Marylebone Roster app's Links modules from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.